TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders

Intercepts: set y = 0 to find the x-intercept; set x = 0 to find the y-intercept.

Solution of a system: the point where the lines cross. It makes both equations true.

How many? Different slopes: one. Same slope, different y-intercept: none. Same line: infinite.

1. What is the x-intercept of $2x + 5y = 20$?

Tip: x-intercept: set $y = 0$

A. (0, 4)

B. (4, 0)

C. (10, 0)

2. What is the y-intercept of $3x + 4y = 24$?

Tip: y-intercept: set $x = 0$

F. (0, 6)

G. (8, 0)

H. (0, 8)

3. Is (2, 3) a solution of $y = 2x - 1$?

> **Tip:** put x and y in

A. No, because $3 \neq 5$

B. Yes, because $3 = 3$

C. Yes, because $2 = 2$

4. What is the slope of $y = -\frac{1}{2}x + 4$?

> **Tip:** $y = mx + b$: m is the slope

F. 4

G. $\frac{1}{2}$

H. $-\frac{1}{2}$